

O U T L I N E S
O F
M I N E R A L O G Y,

T R A N S L A T E D
F R O M T H E O R I G I N A L,

O F
S I R T O R B E R N B E R G M A N,
K N I G H T O F T H E O R D E R O F W A S A,
P R O F E S S O R O F C H E M I S T R Y A T U P S A L, & C.

B Y W I L L I A M W I T H E R I N G, M. D.
M E M B E R O F T H E R O Y A L M E D I C A L S O C I E T Y,
A T E D I N B U R G H.

*Itum est in viscera terræ;
Quasque recondiderat stygiisque admoverat umbris
Effodiunter opes, irritamenta malorum.*

O V I D.

B I R M I N G H A M:
P R I N T E D B Y P I E R C Y A N D J O N E S,
F O R T. C A D E L L, A N D G. R O B I N S O N, L O N D O N;
J. B A L F O U R, A N D C. E L L I O T T, E D I N B U R G H.
M, DCC, LXXXIII.

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F R O M T H E O R I G I N A L,

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K N I G H T O F T H E O R D E R O F W A S A,
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B Y W I L L I A M W I T H E R I N G, M. D.
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*** START OF THE PROJECT GUTENBERG EBOOK OUTLINES OF
MINERALOGY ***

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OUTLINES
OF
MINERALOGY,
TRANSLATED
FROM THE ORIGINAL,
OF
SIR TORBERN BERGMAN,
KNIGHT OF THE ORDER OF WASA,
PROFESSOR OF CHEMISTRY AT UPSAL, &C.
BY WILLIAM WITHERING, M. D.

MEMBER OF THE ROYAL MEDICAL SOCIETY

AT EDINBURGH.

Itum est in viscera terræ;
Quasque reconsiderat stygiisque admoverat umbris
Effodiunter opes, irritamenta malorum.

OVID.

BIRMINGHAM:

PRINTED BY PIERCY AND JONES,

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M,DCC,LXXXIII.

TRANSLATOR'S PREFACE.

The pleasure and instruction I received myself from this excellent little work of Professor Bergman, inspired me with a wish to make it more generally known to others. A system like this, founded upon the constituent principles of things, may be improved, but never can be exploded. English names are given, but the Latin ones of the original are still retained, as an acquaintance with them will enable the reader more readily to consult other authors. Blank spaces are left after most of the species, for the convenience of inserting any new ones that may occur. I have added a few new species, and some notes; the utility of which will be sufficiently obvious. The table of metals, at page 71, and the index at the end, will also, I hope, be considered as useful additions.

BIRMINGHAM,
1st September, 1783.

N. B. The centenary (*centenarius*) of PROFESSOR BERGMAN is equal to 60 Swedish grains, or nearly 63 English grains.

THE AUTHOR'S PREFACE.

In compliance with the request of my learned and amiable friend, the celebrated Mr. Ferber, I transmitted to him a flight sketch of mineralogy, in which the subjects were arranged according to their constituent or component parts. After perusing it, he requested my permission to publish it. At first I thought it better to suppress a work that was so imperfect, especially when I considered the number of analyses that yet remained to be made. He replied, that a perfect method was not yet to be expected in a subject so extensive, but that having once laid a good foundation, I might occasionally make such additions and corrections, in new editions of the work, as future experiments might render necessary. Indeed, I was fully aware, that the system would sooner be rendered perfect, if submitted to the inspection of other more discerning chemists, than if the completion of it rested upon myself only. The different remarks of others, will correct errors, which, by a further attention, I might have amended; but if the interest of science be promoted, no matter by whom.

This little work contains GENERA and SPECIES, except in the appendixes, which, as not properly belonging to my design, contain Genera only.

The GENERA are founded upon the prevalent component parts; the SPECIES upon the diversity of the composition. *Varieties* depend upon external appearances, and therefore are at present omitted.

After this manuscript was sent away, I discovered two species of *stannum fulphuratum* (tin combined with sulphur), one of which contains about forty per cent. of sulphur, the other only twenty. The first has the appearance of aurum musivum; the latter partly resembles *antimonium fulphuratum* (crude antimony), but does not contain antimony. Both are contaminated by a small quantity of copper. I got them from Nerchinskoi in Siberia^[1].

As to the TERRA PONDEROSA (heavy earth), I have long been aware of its great resemblance to calx of lead, and have even lately found a method of precipitating it by the phlogisticated alkaly^[2]; so that I verily believe it to be of a metallic nature, although it has never yet been made into a regulus, and, therefore, I still place it with the earths, until its situation be better ascertained.

If providence allots me life and health, I hope, a few years hence, to republish this imperfect sketch, corrected and enlarged.

OF A
NATURAL SYSTEM
OF
MINERALOGY.

§ 1.

The MINERAL KINGDOM consists of the fossil substances found in the earth. These are either entirely destitute of organic structure, or, having once possessed it, possess it no longer: such are the petrefactions.

§ 2.

It is requisite, for the proper discrimination of fossils, to establish certain characters, whereby they may, at all times, and in all places, be distinguished from one another. The science that teaches these is called MINERALOGY.

§ 3.

As in the vegetable kingdom different methods have been formed upon the roots, the leaves, the flowers, the fruit, &c. so also in Mineralogy many methods may be devised, and there is no doubt of the utility of contemplating inorganic bodies in every point of view; for the more comparisons are multiplied, the more evidently do resemblances or differences appear.

§ 4.

But as the chief object of the science is to render fossils subservient to the uses of man, it is evident that that method must be the best which displays their component parts: for these being well understood, we know what to expect from them; we accommodate our designs to their nature, and spend not our labour and money in vain attempts inconsistent with their inherent qualities.

§ 5.

There is a power implanted by the creator in organized bodies, which, upon the acquisition of proper nutriment, unfolds and evolves the structure which before lay concealed in the fecundated egg or seed. Similar vessels, in each species, absorb, convey, and assimilate the nourishment in the same manner; so that the appearance and structure remain the same, unless peculiar causes prevent the accustomed course of things, and produce monsters: but this very rarely happens. Hence it is that the leading features or the external parts agree with the internal properties, and when judiciously chosen, form sufficient characteristic distinctions.

§ 6.

But the formation of fossils is totally different. Here no system of vessels collects, distributes, secretes or changes the concurrent particles, but they run together by chance, and are solely connected by the power of attraction; they are generally, too, of different kinds, rare and dense, figured and shapeless, admitting of every possible variety. This general view of the subject shews us how little external characters can be depended on; but we shall more particularly consider the principal of these.

§ 7.

Colour varies exceedingly, as does also the size of bodies. We cannot sufficiently wonder at the violence done to nature by the studied separation of earths from stones. The consequence is, that a stone of a certain size must constitute one genus, whilst the same thing, reduced to powder, must be placed under another genus, which shall not be found even in the same class.

§ 8.

Hardness not unfrequently varies even in the same specimen. Soft clay dries in the fire, and at length acquires the hardness of flint. Steatites (*soap-rock*) which may be scraped with the nail, and many other matters harden in the same manner, and that sometimes without any notable loss of weight; so that bodies pass through every different degree of hardness, without any other change of their mixture.

§ 9.

TEXTURE, and external form of the particles, may seem at first sight to depend more upon the constituent parts; but a calcareous particle, globular or shapeless, is found, upon the most scrupulous examination, to possess the same properties as a piece of spar; and in another place I have clearly shewn, that the schirl-like, garnet-like, hyacinthine, twelve-sided, and other figures, are not unfrequently formed by nature out of the same materials^[3]. And if we are liable to deception where so great a difference in external forms exists, what can we expect from less constant external qualities?

§ 10.

Superficial characters are therefore insufficient. They cannot even enable us to distinguish calcareous from other earths, for the effervescence with acids is a chemical mark, and happens, too, in matters of very different natures. To pass over other instances, let him who is able distinguish the *plumbum aeratum* and *plumbum phosphoratum* (§ 182. § 183.) by external appearances only!

§ 11.

But let us not altogether despise external characters: it is of moment to know and mark them well^[4]. They frequently enable the accustomed eye without troublesome trials to acquire a degree of certainty, which wants only a few select experiments to confirm it. Sometimes also the use depends upon external properties, evident to our senses, as the hardness, the colour,

the pellucidity, &c. These therefore may with propriety be joined to those which point out the constituent principles.

§ 12.

Classes, Genera and Species are therefore to be formed upon the internal nature and composition; the varieties upon the external appearances. In such a system both methods conveniently agree.

§ 13.

CRONSTEDT first attempted this method, and with great success; but afterwards the *liquid analysis*, in which the illustrious MARGRAAF took the lead, better opened the internal secrets of nature; so that the excellent work of Cronstedt now appears to contain many errors; these however are not to be attributed to any fault in the author, but to the insufficiency of his experiments. The attempts of Mr. Pott by fusion have long been known; but these however useful in other respects, rather tend to confound than to lay open the component parts of bodies.

§ 14.

In methodizing fossils, *compounds should rank under the most abundant ingredient*. Thus let *a* and *b* represent the component parts; if the former be the heavier, the compound must be placed under the genus of that: but this rule admits of several exceptions.

§ 15.

Thus, the *properties of all ingredients are not of the same intensity*, if I may be allowed the expression; some are more powerful or efficacious, so as to impress the mass with their own genus and character, though forming *less* than half the weight. In such a case the qualities are rather to be considered than the quantity, especially if *b* so far from preponderating hardly ever amounts to half the weight.

§ 16.

Argillaceous earth (*earth of allum*) and magnesia are never found separate, but almost always mixed with other things so that their weight constitutes the smaller part of the mass: therefore if the above rule (§ 14.) was rigorously adhered to, these primitive earths would not be found amongst the Genera, which would doubtless be an absurdity.

§ 17.

The *value* of a thing must likewise be considered. Minerals containing gold or silver must be ranked with those noble metals although they hold three, four, or more times the quantity of heterogeneous matter. Not to mention other examples, pyrites are placed under the genus copper although they contain a much greater quantity of iron. This custom, established with the universal consent of mineralogists, wants indeed a natural foundation, but it seems useful to miners to retain it; and the more so as it is certain that otherwise many minerals would be to be sought for under strange and improper titles.

§ 18.

Lastly, it must be remarked, that *the solid ingredient determines the genus* although the menstruum be greater in quantity. Thus in magnesia vitriolata (*Epſom ſalt*) the earth gives the Generic name, although the vitriolic acid be the more ponderous. The same holds good in gypsum, allum, &c.

CLASSES OF FOSSILS.

§ 19.

Fossils are of four kinds, viz. *saline, earthy, inflammable, and metallic*; hence arise four classes.

§ 20.

Salts, or saline substances are more or less sapid, and when finely powdered dissolve in at least 1000 times their weight of boiling water. They melt in the fire, which for the most part changes or destroys them^[5].

§ 21.

Earths are insipid, not soluble in water in the degree mentioned above (§ 20) though perhaps water in Papin's digester will dissolve some if not all of them, especially if their surface be greatly increased by a previous solution in and precipitation from some other menstruum. In the chain of nature they are by insensible gradation joined to the salts, so as not to be distinguished without artificial limits. Their form is not changed by a moderate heat, nor are they dissipated by a violent one. Their specific gravity is to water, less than 5 to 1.

§ 22.

Inflammable fossils abound with phlogiston, do not unite with water, but when pure dissolve in oils; exposed to the fire, they smoke, generally inflame, are for the most part consumed, and sometimes totally vanish.

§ 23.

Metals when perfect do not dissolve at all in water; only a few of them in oils, and then only when in part deprived of their phlogiston. They are the heaviest of all known substances, the lightest of them weighing more than six times its bulk of water.

They melt in the fire with a shining surface, and in clay vessels the surface is convex.

CLASS I. SALTS.

§ 24.

We begin with the nature and properties of saline bodies, for unacquainted with these our knowledge of other bodies must be exceedingly imperfect. *Native salts* are either *acid, alkaline, neutral, earthy* or *metallic*.

§ 25.

Acids may be distinguished by their proper taste; they effervesce with mild alkalies; and change the blue juices of vegetables and tincture of heliotropium to a red colour^[6].

We are acquainted with many species of acids, but they are hardly ever found pure in the bowels of the earth, nor can we expect to find them so when we consider how soon such powerful menstrua must meet with substances to saturate them. Their great abundance and their properties shew their various and indispensable uses in the œconomy of nature.

§ 26.

As mineralogy treats of those bodies which are found under the surface of the earth, and as acids in an uncombined state are not found there, it would seem proper to exclude them; but the same reason would likewise exclude the primitive earths, some of which have never yet been found pure. Therefore in a system formed upon the component parts of bodies, a short description of the principal of these is not to be dispensed with, although they hardly ever present themselves in a separate state.

§ 27.

Vitriolic ACID. When most concentrated by artificial means its specific gravity is 2, 125. When pure, has neither colour nor smell. Cold sometimes though very rarely concretes it into a solid form; it may be coagulated by nitrous air. This as well as the other acids is best known from the compounds it forms with other substances.

Mr. VANDELLI^[7] says that it is sometimes mixed with the streams from the hills in the neighbourhood of Sienna and Viterbo, raised no doubt by subterranean fires; but in general it is united to alkalies (§§ 44, 47, 50,) to earths (§§ 58, 59, 63, 67,) to metals (§§ 69, 70, 72, 73,) or to phlogiston (§§ 134, 136.)

Phlogifticated vitriolic ACID (volatile vitriolic acid) is frequently thrown out by the craters of volcanoes; its smell suffocating and penetrating. The union to phlogiston and the matter of heat gives it an aerial form, but does not prevent its union with water.

§ 28.

Nitrous ACID is by some excluded from the fossil kingdom, because they suppose it to be produced from the putrefaction of organic bodies. But these bodies when deprived of life are again received amongst the fossils, from whence their more fixed parts were originally derived.

In the most concentrated state that art can procure it, its specific gravity is 1, 580. Colourless when pure; but its strong attraction to phlogiston renders particular management necessary to procure it so^[8]. With different proportions of phlogiston it forms phlogifticated acid and nitrous air.

It has never as far as I know been met with disengaged, unless perhaps in water precipitated out of the atmosphere, but is found united to alkalies (§§ 45, 47, 51) or to earths (§§ 60, 64.)

§ 29.

Muriatic ACID (spirit of salt) is found in great quantity at and under the surface of the earth. The strongest prepared by art hardly attains a specific gravity of 1, 150. It has a very peculiar and volatile smell. Deprived of its

superfluous water it assumes an aerial form, for phlogiston seems to be one of its constituent parts^[9].

It has never been found uncombined (unless perhaps like the nitrous acid in water precipitated from the atmosphere^[10]^[11]) but united to alkalies (§§ 46, 49, 52), to earths (§§ 61, 65), or to metals (§§ 74, 161, 175, 191).

§ 30.

Fluor ACID, is obtained by art; its specific gravity never exceeds 1,500, it is very volatile. Its vapours when hot, corrode glafs; and meeting with moisture generate, or at least deposit filiceous earth. When deprived of its superfluous water it assumes an aerial form^[12]. It has never been found disengaged, but united to calcareous earth forming sparry fluor^[13] (§ 96) and if I am not mistaken it enters into the composition of filiceous earths.

§ 31.

Arsenical ACID, dry; prepared by art; specific gravity 3,391; fusible and fixed in the fire, until it acquires from the matter of heat so much phlogiston as is necessary to convert it into white arsenic. In a moist air it deliquesces.

It is not found uncombined, but united to calx of cobalt (§ 228), and also to phlogiston, forming a brittle arsenical metal (§ 220), and its calx (§ 222).

§ 32.

Molybdæna ACID. This is very probably of metallic origin, though it does not yet appear to which metal it belongs. Seeing that arsenic, a brittle metal, by dephlogistication only is changed into an acid, different from all other acids, it is not improbable that other metals may have an acid basis, although their phlogiston adhering more strongly has not yet been completely separated.

How this substance may be obtained by art does not belong to this place to describe^[14]; but that the acid got from Molybdæna has a metallic nature, and as yet has not been perfectly freed from phlogiston, is probable from the following considerations. 1, Its taste is acid and at the same time metallic. 2, Microcosmic salt and borax are coloured by it, and these salts are hardly

coloured by any thing but metallic calxes. 3, Its decomposition by means of the phlogisticated fixed alkaly, which always indicates the presence of a metal. 4, Its concrete form, and not deliquescing, analogous to white arsenic. 5, Its specific gravity 3,460. And very lately M. HIELM by my persuasion attempted the reduction and obtained a regulus, seemingly different from every other metal, but not yet sufficiently examined.

§ 33.

An acid conjoined to the calx ponderosa (*ponderous calx or lime*) is nearly allied to the preceding, but dropped into lime water produces a different compound, though in a number of other circumstances these two acids agree. I apprehend that this is likewise of a metallic nature.

§ 34.

Phosphoric ACID, evidently exists in the animal kingdom,^[15] much more plentifully in the vegetable, but in the fossil very rare. Mr. J. G. GAHN first detected it united with lead;^[16] but probably it may be found in many other fossils. It is fusible in the fire. Its specific gravity when deprived of water 2,687.

§ 35.

Boracic ACID, (*acid of borax, or sedative salt.*) Many people still think this to be an artificial production, but not long since Mr. HOEFER^[17] found it in a lake near Sienna in the great duchy of Hetruria, and it has long been known to be united to the fossil alkaly in native borax. It acts like an acid, though very feebly. It melts in the fire and volatilizes with water. Its specific gravity is 1,480.

§ 36.

Amber ACID, is a concrete salt obtained from amber; it acts like a feeble acid. It is yet doubtful whether amber be of vegetable origin; many reckon it fossil.

§ 37.

Aerial ACID (fixed air) is not only combined with water but with many other fossil substances, as alkalies (§§ 54, 56), earths (§§ 62, 66), and with some metals (§§ 71, 183, 192, 217, 234, 243). It floats uncombined in the atmosphere. Its specific gravity 0,0018^[18].

§ 38.

ALKALIES are known by their peculiar lixivial taste, by their vehement attraction to acids, and by their changing the blue colours of vegetables to a green. In a pure state, as was before observed of acids, their attraction to other substances is so strong that they cannot long remain uncombined; and if other acids were wanting, the aerial acid, every where present in the atmosphere, would unite with them: therefore they are always found in a state of combination, unless prepared by art.

§ 39.

New acids are daily detected, but no additions have been made to the three species of alkaly long since known.

§ 40.

Vegetable fixed ALKALY, deprived of every acid is not found on the face of the earth; but it is sometimes met with in combination with the vitriolic acid (§ 44) or the muriatic (§ 46), generally with the nitrous, (§ 45) rarely with the aerial (§ 54).

§ 41.

Fossil fixed ALKALY is only found in combination with acids, rarely with the vitriolic (§ 47) or nitrous (§ 48), principally with the muriatic (§ 49) or aerial (§ 55).

§ 42.

Volatile ALKALY is frequently found in clays, doubtless in a mild state, for the help of art is required to render it caustic. It is also found united to the vitriolic (§ 50) and the muriatic acids (§ 52.)

§ 43.

ACIDS united to *alkalies* form NEUTRAL SALTS. These dissolved in water are no ways disturbed by the addition of an alkaly, and generally by evaporation concrete into crystals. If by proper tests they shew neither acid nor alkaline properties they are said to be *perfect* neutrals, but *imperfect* when from defect in quantity or strength of one ingredient the peculiar properties of the other more or less prevail.

We now proceed to consider the native salts of both kinds.

NEUTRAL SALTS.

§ 44.

Alkali vegetabile vitriolatum (tartar of vitriol) seldom occurs spontaneously, unless where tracts of wood have been burnt down.

§ 45.

Alkali vegetabile nitratum (common nitre) forms upon the surface of the earth where vegetables, especially when mixed with animal substances, putrify. The alkaline basis previously exists in the plants^[19], but the origin of the acid is not so well understood: whether it lies concealed in the vegetable acid, and by means of the putrefactive process sufficiently dephlogisticating it, is evolved; or whether the purer part of the atmospheric air contains nitrous acid *fully* saturated with phlogiston, which^[20] upon the alkali being separated by the putrefaction is attracted and extricated by it, and upon losing its inflammable principle assumes its accustomed form. Nature perhaps operates in both ways; the latter however seems clearly confirmed by a very remarkable experiment (§ 60.)

As nitre is annually produced in large quantities, it cannot but sometimes be found in springs or wells, as has been observed at Berlin^[21], London^[22], and elsewhere^[23]. Sometimes it abounds in such quantities that flesh boiled in these waters turns red.

§ 46.

ALKALI *vegetabile falitum* (digestive falt) is sometimes though rarely met with; generated perhaps by the destruction of animal and vegetable substances.

§ 47.

ALKALI *minerale vitriolatum* (Glauber's falt) is sometimes found in waters. Some of the lakes in Siberia and Astracan contain it, and many springs in other places.

§ 48.

ALKALI *minerale nitratum* (cubic nitre) rarely occurs but where maritime plants putrify.

§ 49.

ALKALI *minerale falitum* (common falt) plentiful every where as well in the earth, where it forms strata more or less thick (fal gem), as also dissolved in springs and lakes, and in the sea. (sea falt.)

§ 50.

ALKALI *volatile vitriolatum* (vitriolic ammoniac) is scarcely found any where but in places where the phlogisticated fumes of vitriolic acid arise from burning sulphur, and in putrid places are absorbed by the volatile alkaly.^[24] Thus at Fahlune the acid vapour from the roasted minerals produces this falt in the necessary houses. It is sometimes also formed in the craters of volcanoes.

§ 51.

ALKALI *volatile nitratum* (nitrous ammoniac) is generally found along with common nitre.

§ 52.

ALKALI *volatile falitum* (fal ammoniac or common ammoniac.) I have examined some from Vesuvius, and some from the Solfaterra near Naples.

The salts hitherto enumerated are perfect neutrals, those which follow are imperfect (§§ 53, 56.)

§ 53.

ALKALI FOSSIL, only in part saturated with a peculiar acid is called tinkal; after depuration, borax. It is dug out of the earth in the kingdom of Thibet^[25]. Borax takes nearly an equal weight of acid before the alkaline properties entirely disappear^[26].

I believe no one has yet found the acid of borax united either to the vegetable or volatile alkalies.

§ 54.

ALKALI VEGETABILE *aeratum* (mild vegetable alkaly) is hardly ever found native, unless in the neighbourhood of woods destroyed by fire.

In the year 1774, at Douai in Flanders, a spring was discovered surrounded by a wall, whose waters, besides other impregnations, contained 11 grains of vegetable alkaly in a pint^[27].

§ 55.

ALKALI MINERALE *aeratum* (mild fossil alkaly, natron, the nitre of the ancients) is found plentifully in many places, particularly in Africa and Asia, either concreted into crystallized strata, or fallen to a powder; or efflorescing on old brick walls, or lastly, dissolved in springs. It frequently originates from decomposed common salt. I am not ignorant that the acid of common salt adheres strongly to its basis so as not to be expelled by fire; but perhaps the vicissitudes of the atmosphere continually acting for ages, may be more powerful. In immense plains covered over with this alkaly, scarcely any common salt is found upon the surface, but the deeper you dig the more

it is contaminated by it, the common falt being yet undecomposed for want of access of air.

§ 56.

ALKALI VOLATILE *aeratum* (mild volatile alkaly) has been found in pump waters in London^[28], in Lauchstadt^[29], at Frankfort on the Mayne^[30], and copper immersed therein is said to have been dissolved into a blue liquor.

The three alkalies mentioned above as saturated with aerial acid, differ greatly from caustic alkalies, in the mildness of their taste, in their property of crystallizing, and in their effervescing with acids which expel the aerial acid, but they still change vegetable blues to greens, though not so powerfully as the caustic alkalies do. Therefore, although the subtil aerial acid in other respects gives them neutral properties, yet in this it does it but imperfectly.

§ 57.

The compounds of earths and acids which possess solubility mentioned at § 20, are decomposed and precipitated by mild, but not by phlogisticated alkalies.

§ 58.

TERRA PONDEROSA *vitriolata*, (heavy spar, marmor metallicum, calc) is placed with the earths (§ 89.) Terra ponderosa *nitrata* i. e. terra ponderosa united to the nitrous acid, perhaps exists somewhere, but has never been met with; neither has the terra ponderosa united to the *aerial* acid, yet been found^[31]. Terra ponderosa *salita* i. e. terra ponderosa with the *muriatic acid* Mr. HIELM says^[32] is dissolved in the waters of the lake Vettern and its neighbourhood.

§ 59.

CALX *vitriolata* (gypsum, selenite) is not only found dissolved in various waters, but also in many places forms immense strata. It is placed by all mineralogists amongst the earths, but I think improperly. When burnt it generates heat with water, but in a less degree than lime does.

§ 60.

CALX *nitrata* (nitre of lime; terrene nitre) is sometimes found in water, but very sparingly. It is said that the chalk hills in some parts of France become spontaneously impregnated with nitrous acid, which may be washed out, and after a certain time they will become impregnated with it again.

§ 61.

CALX *Salita* (fixed ammoniac) occurs very frequently in waters.

§ 62.

CALX *aerata* (marble, limestone, chalk, spar) is very commonly found dissolved in waters in consequence of an excess of the aerial acid. When it greatly abounds, the water is said to be hard (*cruda*). By boiling, or by evaporation, it deposits streaks or crusts of calcareous matter.

Calx aerata is not soluble in water without an excess of the subtil acid, and therefore might properly be referred to the earths (§ 21).

§ 63.

MAGNESIA *vitriolata* (Epsom salt) is not unfrequent in the waters of England, Bohemia, and other countries. This salt is presently decomposed by lime water, which circumstance readily distinguishes it from the alk. min. vitriol. or Glauber's salt.

§ 64.

MAGNESIA *nitrata* (magnesia and nitrous acid) is usually found together with nitre.

§ 65.

MAGNESIA *falita* (magnesia and muriatic acid) is found dissolved in various waters, but plentifully in sea water, to which it gives a disagreeable bitterness.

§ 66.

MAGNESIA *aerata* (common magnesia) with an excess of aerial acid it becomes soluble in cold water, otherwise it is scarce soluble at all, and therefore should be classed with the earths. (§ 21.)

§ 67.

ARGILLA *vitriolata* (alum) is sometimes spontaneously generated by the decomposition of pyrites lodged in clay, or in argillaceous schistus.

It is found in a spring at Steckenitz in Bohemia^[33], in East Bothnia and elsewhere. What is commonly called *plumose* alum is not a saline substance.

ARGILLA (clay) united to the *nitrous*, *muriatic*^[34], or aerial acids has not to my knowledge hitherto been found in any waters.

METALLIC SALTS.

§ 68.

The native salts belonging to this division, may be distinguished by the phlogisticated alkali which precipitates them all. The few which have saline properties (§ 20.) we shall mention here, referring the rest to the mineralized metals.

§ 69.

CUPRUM *vitriolatum* (vitriol of copper, blue vitriol) is found in the mines of Herregrund, Fahlun, and others which contain copper pyrites.

§ 70.

FERRUM *vitriolatum* (vitriol of iron, green vitriol) is formed from the decomposition of the more common pyrites.

§ 71.

FERRUM *aeratum* (iron with aerial acid) dissolved by an excess of acid in the lighter chalybeate waters.

FERRUM *nitratum*, and *fulitum* (iron with nitrous and muriatic acids) have never yet been found native.

§ 72.

NICCOLUM *vitriolatum* (vitriol of Nickel) sometimes exists from the decomposition of sulphureous ores of Nickel.

§ 73.

ZINCUM *vitriolatum* (vitriol of zinc, white vitriol) is sometimes, though rarely, produced By the decomposition of pseudogalena, or black Jack, because this substance does not very readily decompose spontaneously.

§ 74.

^[35]MANGANESIIUM *salitum* (manganese united to muriatic acid) exists in some waters Mr. HIELM says.

Whether manganese be ever united to waters like iron, by means of an excess of aerial acid, we know not.

TRIPLE SALTS.

§ 75.

The compound salts hitherto enumerated are such as are composed of two ingredients only; but sometimes three or more are so united as not to be separated by crystallization. The vitriols that we are acquainted with are hardly ever pure, and two or three of them sometimes are joined together.

Sometimes likewise it happens that neutral salts join earthy salts, and earthy salts metallic ones. I generally distinguish compound salts according to the number of their principles, whether the same acid be joined to several bases, or the same basis to different acids; or lastly, whether several menstrua and several bases are joined together. Hence arise salts triple, quadruple, &c. which the diligence of after times must illustrate. I subjoin the most remarkable examples of triple and quadruple native salts which have occurred to me.

§ 76.

ALKALI MINERALE SALITUM (common salt) contaminated by *magnesia falita*. The common salt when pure does not deliquesce, but this degree of purity is seldom found, and in the native fossil production (*sal gem*) never.

§ 77.

MAGNESIA *vitriolata* (Epsom salt) contaminated by *ferrum vitriolatum*^[36] (vitriol of iron.)

§ 78.

ARGILLA *vitriolata* (alum) *native*, contaminated by *vitriol of iron*. In the aluminous schistus it sometimes effloresces in a feathery form. Is this the plumose alum of the ancients?

§ 78*.

ARGILLA *vitriolata* (alum) *native*; contaminated by sulphur and vitriolic acid.

At the places about Wednesbury and Billton, in Staffordshire, where the coal pits are on fire, this substance sublimes to the surface, and may be collected in considerable quantity during dry or frosty weather. I cannot be certain that this is a true chemical union, but the eye cannot distinguish the parts. Perhaps the sulphur volatilizes the alum and so becomes intimately mixed with it. The excess of vitriolic acid keeps it in a deliquescent state.

I believe a similar compound substance sublimes at the Solfaterra near Naples. W.

§ 79.

ARGILLIA *vitriolata* (alum) *native*, contaminated by *vitriol of cobalt*. In the mines of Herregrund and Idra this may be seen, shooting out into long slender filaments. Perhaps this is the *trichites* of the Greeks. Dissolved in water it immediately betrays the presence of vitriolic acid, upon the addition of terra ponderosa salita (muriatic acid saturated with heavy earth.) By the addition of phlogisticated alkali a precipitate of cobalt is thrown down, which makes a blue glass with borax or microcosmic salt.

§ 80.

CUPRUM *vitriolatum* (vitriol of copper) contaminated by *iron*.

§ 81.

FERRUM *vitriolatum* (vitriol of iron) contaminated by *nickel*.

§ 82.

CUPRUM *vitriolatum* (vitriol of copper) and vitriol of iron contaminated by *zinc*. Such is found at Fahlune.

CLASS II. EARTHS.

§ 83.

Before we can understand the nature of earths, we must know their component parts. Those earths which cannot be further decomposed we call *primitive*, and those which consist of two or more of these intimately united, *derivative*. By this union we do not mean a mere mechanical diffusion, at least not such as can be distinguished by the eye, as is the case in stones, (*faxa*.)

§ 84.

It is evident that the primitive earths will constitute so many natural Genera, and different mixtures of these the Species.

They who would make several Genera out of one primitive earth, must separate the glassy, red, white, horny silver ores, and other different compositions into as many Genera, or else act inconsistently with their own principles.

§ 85.

At present we only know five primitive earths. They who reckon fewer, rest their opinions upon fanciful metamorphoses unsupported by faithful experiments^[37]. As experiments teach us that there are five primitive earths, it is evident that the Species arising from the mixture of these cannot exceed twenty-four, viz. 10 double (consisting of two earths) 6 triple, 3 quadruple, and the 5 primitive.

Although these different mixtures are possible, and probably do exist, they have not yet been all found. The natural compositions of acids with the earths, forming substances not soluble in 1000 times their weight of boiling water, and which may be called saline earths, must be added to the species, as they are certainly chemical combinations.

§ 86.

The primitive earths hitherto detected are,

TERRA PONDEROSA, or	heavy earth.
CALX,	calcareous earth.
MAGNESIA,	magnesia.
ARGILLA,	argillaceous earth.
TERRA SILICEA,	siliceous earth.

And we must believe these to be primitive, until it shall appear by proper experiments that they may be separated into others still more simple, or changed into one another by art.

These are first to be considered in their greatest simplicity and purity, although nature never presents us with such, nor can they even by art be rendered absolutely free from all heterogeneous mixture. Water and aerial acid readily unite with the four first, and when expelled by fire, a little of the matter of heat is added, and remains until driven out by a more powerful attraction. But in this state they possess a degree of purity not to be attained by any other known method. Therefore it is necessary to examine them when sufficiently burnt in order to distinguish better what properties depend upon adhering heterogeneous matters.

HEAVY EARTH,
OR
TERRA PONDEROSA.

§ 87.

To obtain this as pure as possible, the *spathum ponderosum* § 89 (heavy spar) must be reduced to a fine powder, and with equal parts of fixed alkaly and charcoal dust roasted for an hour in a covered crucible. Powder the mass, and add nitrous or muriatic acid diluted until all effervescence ceases, and the liquor be sensibly acid. To this liquor add mild fixed alkaly, and the heavy earth will be precipitated in a mild state. If the acids or the alkaline salt contain any vitriolic acid, the heavy spar will immediately be regenerated. What remains undissolved by the acid is heavy spar, not decomposed. The process may be repeated upon this, but the product will then contain some martial earth and some clay from the crucible, therefore the first part will be the most pure.

§ 88.

TERRA PONDEROSA *aerata*, (heavy earth) has a specific gravity of 3, 773^[38]. 100 parts of it contain about 28 of water, 7 of aerial acid, and 65 of pure earth. It effervesces with acids: with the vitriolic acid forms heavy spar, not soluble in water; with the nitrous and muriatic acids, it yields crystals, not very readily soluble; but with the vegetable acid the crystals deliquesce.

When free from all contamination of acid or alkaly it scarcely melts in the fire, but loses $\frac{35}{100}$ of its weight. When united with the matter of heat, (i. e. rendered caustic) it dissolves in 900 times its weight of water; and when this solution is exposed to the atmosphere, a cream or crust separates at the top,

which effervesces with acids. After burning, it unites to acids without effervescence; but heat is produced, and the union is more tardy than when it is in a mild state^[39].

When caustic, it expels the volatile alkaly from sal ammoniac, and forms a hepar with sulphur, the watery solution of which is but imperfectly decomposed by the nitrous or muriatic acids, upon account of the remarkable attraction betwixt this earth and the acid of the sulphur, which it even takes from the vegetable alkaly^[40].

When we compare these properties with those which belong to common calcareous earth, mentioned at (§§ 92, 93), we shall readily see wherein they agree, and wherein they differ.

§ 89.

TERRA PONDEROSA *vitriolata* (heavy spar) is full four times as heavy as an equal bulk of water. It dissolves entirely, though sparingly, in concentrated boiling vitriolic acid, but the addition of a single drop of water occasions a precipitation. The same thing happens to gypsum; but that requires much less acid to dissolve it, and the precipitation is made more slowly. If the heavy spar contained any sulphur, it must certainly have appeared when the whole was dissolved, but I never could find any thing like it.

CRONSTEDT, Min. § 18. 2.

Marmor metallicum drusicu § 19 C. Ponderous Spar.

§ 90.

TERRA PONDEROSA *vitriolata*, impregnated with bitumen, and mixed with gypsum, alum, and siliceous earth.

CRONSTEDT Min. § 24. *Lapis hepaticus*. Liver Stone.

A nucleus of this kind, taken out of a piece of alum ore from Andrarum in the province of Skone, yielded, in 100 parts, by analysis, 33 of siliceous earth, 29 of caustic heavy earth, earth of alum about 5, and quick-lime from

3 to 7, besides the water and vitriolic acid. By calculation it appears, that these bases, together with vitriolic acid enough to saturate them, ought to weigh 71, which, with the addition of 33, exceeds the amount of the original 100. This increase points out the difference of a mass newly crystallized, and of one carefully dried.

§ 91.

When we consider that the terra ponderosa was altogether unknown before the year 1774, and that many mineralogists are even now unacquainted with it, we cannot wonder that we know so few species of it. I have scarce a doubt but the *terra ponderosa aerata* may be found mixed with other earths in many specimens, when they come to be examined by chemical means more accurately than they could be heretofore. (See notes to §§ 58 and 88.)

CALCAREOUS EARTH,

OR

CALX.

§ 92.

As calcareous earth united to the aerial acid is found native, it requires but little trouble to have it pure. Let selected pieces of chalk, reduced to fine powder, be repeatedly boiled in pure water: this dissolves any calx or magnesia falfa which it may contain. This done, it holds no heterogeneous matter but what mechanically adheres to it, the quantity of which is generally extremely small. If we desire to be free from this likewise, dissolve the washed chalk in distilled vinegar, precipitate with volatile alkali, and after washing the precipitate well, dry it.

§ 93.

The specific gravity of calcareous earth thus purified, is 2,720. 100 parts of it contain about 34 of aerial acid, 11 of water, and 55 of pure earth.

Acids unite with it effervescing, and a centenary (centenarius) excites about 22 degrees of heat. The vitriolic acid forms gypsum, difficult to dissolve, (§ 59). The nitrous and muriatic acids form deliquescent salts (§§ 60, 61), and the acetic acid permanent crystals.

Pure calcareous earth does not melt in the fire, but loses $\frac{45}{100}$ of its weight. It dissolves in 700 times its weight of water, generating heat^[41]. Acids dissolve it, producing from a centenary 252 degrees of heat, but without any effervescence. This last circumstance may be best observed by immersing the burnt earth in water, to dissipate a part of the heat, which would otherwise make the acid boil. The water likewise expels the

atmospheric air from the pores of the lime. In this situation, if nitrous or muriatic acid be poured upon it, and if it was previously well burnt, no effervescence will take place. The solution proceeds slowly^[42], but the saturation becomes as perfect as if the calcareous earth had been in a mild state. This burnt earth, or lime, expels the volatile alkaly from sal ammoniac in a caustic state, and it dissolves sulphur; but this compound is separated upon the addition of any acid, even the aerial.

§ 94.

Amongst the native Species of this genus, we must first mention the *CALX aerata* (marble, limestone, chalk) which constitute immense strata. Its chief properties are enumerated above (§ 92). It is very rarely found entirely free from iron, which exists even in the purest Icelandic spar, and indeed in almost every fossil production; upon which account only the more remarkable impregnations with iron will be noticed in the following pages.

CRONSTEDT Min. §§ 5–12.

§ 95.

CALX aerata (calcareous earth *mild*), with more or less *petroleum*. It effervesces with acids, and dissolves; with the vitriolic acid frequently turning brown. Is fœtid when heated or rubbed. The oil is not in sufficient quantity to be collected, by distillation, in drops; it only fouls the inside of the vessels, unless a very great quantity be operated upon. In an open fire the colour presently vanishes, from the petroleum drying up. It generally contains a portion of martial clay.

CRONSTEDT Min. §§ 22, 23. *Lapis fuillus*. Fœtid stone.

§ 96.

CALX fluorata (calcareous earth and *fluor acid*), when pure, is wholly soluble in nitrous and muriatic acids. Exposed to heat, below ignition, it emits a phosphorescent light. Fluor acid, dropped into lime water,

precipitates a powder which has all the properties of the calx fluorata. It is sometimes, but not always, contaminated by a small proportion of filiceous earth and muriatic acid.

CRONSTEDT Min. §§ 97–101. *Sparry fluor*. Blue John.

§ 97.

CALX (calcareous earth) saturated with a *peculiar acid*, perhaps of a metallic nature (§ 33). In acids, particularly in the muriatic, it assumes a remarkable yellow colour, but is not very soluble.

CRONSTEDT Min. § 210. Lapis ponderosus. Tungsten.

§ 98.

CALX *aerata* (calcareous earth *mild*), contaminated by a small proportion of *magnesia falita*.

Magnesia.

§ 99.

CALX *aerata* (calcareous earth *mild*) contaminated by *clay*.

Argillaceous.

§ 100.

CALX *aerata* (calcareous earth *mild*), contaminated by *filiceous earth*.

Siliceous.

§ 101.

CALX *aerata* (calcareous earth *mild*), contaminated by *clay* and *filiceous earth*. (See § 115.)

CRONSTEDT Min. §§ 25. 28. Calcareous Marle.

§ 102.

CALX *aerata* (calcareous earth *mild*), contaminated by *iron* and *manganeſe*. Martial.

CRONSTEDT Min. §30. See alſo §203. *Hæmatites*.

103.

There can be no doubt that the four firſt (§§ 94–97.), if not the laſt (§ 102), are genuine and diſtinct ſpecies; there is ſome difficulty as to the reſt, dependent, perhaps, only upon mechanical mixtures. If the heterogeneous matters can be diſcerned by the eye, we cannot heſitate to refer the ſubſtance to the *faxa* (ſtones); but in theſe the eye cannot diſcern them. Moreover, we know that the earths have a mutual attraction to each other, and form combinations more intimate than mechanical ones. Earth of alum, precipitated by a cauſtic alkali, and thrown into lime water, preſently loſes its pellucid and ſpongy texture, turns white, and condenſes, abſorbing the lime from the water, and forming an union not to be ſeparated but by chemical means.

From theſe conſiderations, I dare not venture to exclude doubtful ſpecies.

We ſay a thing is *contaminated* by another, when the mixture is of the mechanical kind; but when things are joined by the ſtronger power of attraction, we ſay they are *united*.

MAGNESIA.

§ 104.

Magnesia, called in the dispensatories, and by apothecaries *magnesia alba*, is a precipitation from its union with vitriolic acid, called Epsom salt. If this earthy precipitate be wanted in the greatest degree of purity, the Epsom salt must be taken crystallized, and well depurated, dissolved in distilled water, and precipitated by volatile alkali. Let the liquor be boiled for a few minutes, in order that what is kept in solution by the aerial acid may subside.

§ 105.

Magnesia, thus obtained, has a specific gravity of 2,155. 100 parts of it contain about 25 of aerial acid, 30 of water, and 45 of earth^[43]. It dissolves in acids, with a violent effervescence, but without heat. It again forms Epsom salt, with the vitriolic acid; with the nitrous acid it crystallizes, but the crystals are deliquescent; with the muriatic and vegetable acids it does not crystallize, and after drying, greedily attracts moisture from the atmosphere.

It does not melt in a moderate heat, but loses $\frac{55}{100}$ of its weight, and then has no attraction for water; dissolves slowly, even in acids, and that without effervescence, but with some degree of heat. After calcination, it expels the volatile alkali from sal ammoniac, and unites to sulphur, though very feebly.

MAGNESIA *aerata* (common magnesia) is never found native and unconnected, unless in waters, when it is dissolved by an excess of aerial acid. (§ 66.)

§ 106.

MAGNESIA *aerata* (common magnesia) united with *filiceous* matter. This effervesces with acids, and not unfrequently strikes fire with steel.

§ 107.

MAGNESIA intimately united with *filiceous* matter. The soluble part is slowly taken up by acids, without effervescence.

CRONSTEDT Min. §§79–83. and perhaps § 102–105 also; but I have not yet submitted the above to the liquid analysis.

Soaprock. Serpentine.

§ 108.

MAGNESIA united to *argillaceous*, *filiceous*, and *pyritical* matters.

M. MONNET discovered this, and the next species.

§ 109.

MAGNESIA united to *argillaceous*, *filiceous*, and *pyritical* matters, and likewise contaminated by petroleum.

This species resembles aluminous schist, but upon examination is found to contain more magnesia than clay.

§ 110.

All the species, except the first, are more or less contaminated by iron, but they do not owe all their colour to this substance. The green colours altogether vanish during ignition, and leave only a white opaque mass.

ARGILLACEOUS EARTH,
OR
ARGILLA.

§ 111.

By earth of alum (argilla) I do not mean common clay, which is never free from filiceous matter, but a pure clay, unmixed, at least, with any other earth. It may be readily obtained by dissolving Roman or roach alum in distilled water, filtering, and precipitating by mild volatile alkaly.

§ 112.

The specific gravity of this pure clay, or earth of alum, is 1,305. It dissolves in acids, with a little effervescence. With the vitriolic acid it forms alum; with the nitrous, muriatic and vegetable acids, deliquescent salts.

When dry, it absorbs water greedily, becomes soft, and, with a due quantity of water, gains such a tenacity, that it may be moulded at pleasure. This mass contracts greatly in the fire, from whence arise numerous cracks; and with a due degree of heat, it becomes hard enough to strike fire with steel. By this burning it loses its glutinous tenacity, and the water is excluded by the approach of the particles; nor does it again assume its former properties, but by solution and precipitation.

It may be dissolved in the dry way, by means of fixed alkaline salt, as well as in the liquid way, by acids. The vitriolic acid is better than the others for this purpose, because more easily concentrated.

Earth of alum neither dissolves sulphur, nor decomposes sal ammoniac.

§ 113.

ARGILLA (argillaceous earth) united to *filiceous* matter only.

CRONSTEDT Min. §78. *Argilla porcellana*. Porcelain clay.
Pipe clay.

I never examined any clay which did not contain a large quantity of filiceous earth; generally more than half its weight^[44].

§ 114.

ARGILLA (argillaceous earth) united to *filiceous* and *irony* matter.

CRONSTEDT Min. §§ 87 and 90. Bole. Dye-earth.
Clay.

§ 115.

ARGILLA (argillaceous earth) united to *filiceous* and *calcareous* matter.

CRONSTEDT Min. §25. *Marga argillacea*. Marle.

§ 116.

ARGILLA (argillaceous earth) united to *filiceous* earth and *magnesia*.

CRONSTEDT Min. §§ 84, 4. B. Terra lemnia.

Its component parts resemble those of talc, but differ in their proportions, and are also less intimately united.

§ 116*.

ARGILLA (argillaceous earth) united to *filiceous*, *calcareous*, and *magnesia* earths.

Lithomarga.^[45] CRONSTEDT Min. § 84. A.
Stone marrow.

§ 117.

ARGILLA (argillaceous earth) contaminated by *vegetable alkaly* and *fulphur*, or at least by the acid of fulphur.

CRONSTEDT Min. § 124. 2. b. *Minera aluminis romani*.
Alum ore.

It certainly contains vitriolic acid^[46], and perhaps, also, a small portion of fulphur. The vegetable alkaly sufficiently shews its volcanic origin.

§ 118.

ARGILLA (argillaceous earth) contaminated by *filiceous* matter, *pyrites*, and *petroleum*.

CRONSTEDT Min. § 124. 2. c. *Schistus aluminaris*^[47].
Alum slate.

§ 119.

ARGILLA (argillaceous earth) intimately united with *less than half its weight of filiceous* earth, and a small quantity of *mild calcareous* earth.

CRONSTEDT Min. §§ 43–48. Gemma.

The Gems suffer no change under the blowpipe, with fossil fixed alkaly, but are dissolved by microcosmic salt and borax.

To this head belong	<i>Rubinus</i> , the	ruby;
	<i>Saphirus</i> ,	sapphire;
	<i>Topazius</i> ,	topaz;
	<i>Smaragdus</i> ,	emerald.

The *tourmaline* holds a kind of middle place betwixt the gems and the fcherle. The colour, in all of them, is owing to iron.

§ 120.

ARGILLA (argillaceous earth) intimately united to *half its weight of filiceous earth* (or more), and a little *mild calcareous* earth. Scherle.

CRONSTEDT Min. §§ 68–71. *Granatus et Bafaltes*, which I call *Scherle*.

The remote varieties of these are easily distinguished, the near ones difficultly.

§ 121.

ARGILLA (argillaceous earth) loosely united to half its weight, or more, of *filiceous* earth, and a little *calcareous* earth.

CRONSTEDT Min. §§ 108–112. *Zeolithus*. Zeolite.

There is a great affinity betwixt this and Scherle; but in the zeolite, the component parts cohere so loosely, that acids attach and separate them without their being previously treated with alkalies; but this is not the case with the fcherles.

Zeolite, contaminated by magnesia, I have not yet examined.

§ 122.

ARGILLA (argillaceous earth) intimately united to a large proportion of *filiceous* earth, and a small proportion of *magnesia*.

CRONSTEDT Min. §§ 93–96. *Mica. Talcum.* [\[48\]](#)Glimmer. Talc.

SILICEOUS EARTH,
OR
TERRA SILICEA.

§ 123.

This, like the other primitive earths, is seldom found pure. In order to have it so, reduce clear quartz crystals into powder; melt it with four times the weight of fixed alkali; dissolve the whole in water; precipitate by a large quantity of strong acid; carefully wash and dry the precipitate.

The acid must be used in a superfluous quantity, that any other earths contained may be dissolved.

§ 124.

The specific gravity of this earth, is 1,975. The particles, when first precipitated, occupy, in water, at least twelve times the space that they do when dried; so that, when sufficiently fine, they may remain suspended therein; nay, when vehemently heated in a close vessel, they may be dissolved. No acid, except that of fluor spar (§ 30) has any action upon this earth. Fixed alkalies unite with it in the liquid way, but in the dry way they seize it with great vehemence, and convert twice their weight of it into a permanent transparent glass. Such is its affinity to alkalies, that it imparts to clay, which is always loaded with it, the power of separating some of the acid from nitre and common salt. When pure, it is refractory in the fire.

Although siliceous earth is not altogether simple, yet, in mineralogy, it must be considered as primitive, until decisive experiments shew us from which of the preceding earths it is derived^[49].

§ 125.

TERRA SILICEA (filiceous earth) united to very small quantities of *calcareous* and *argillaceous* earth.

CRONSTEDT Min. § 51. *Quartzum*. Quartz.

§ 126.

TERRA SILICEA (filiceous earth) united to *argillaceous* earth.

CRONSTEDT Min. § 58. *Calcedonius*. Chalcedony.

And perhaps the *Opal*. The *Hydrophanus* is only a variety of these.

Whether the *carnelian*, and other *filiceæ*, of finer or coarser texture, belong to this or the preceding species I cannot yet determine with certainty.

§ 127.

TERRA SILICEA (filiceous earth), united to an *argillaceous* and highly *martial* earth.

CRONSTEDT Min. §§ 64, 65. *Jafpis*. Jasper.

§ 128.

TERRA SILICEA (filiceous earth), loaded with *martial* earth.
Martial.

CRONSTEDT Min. § 53.

This species is often called jasper, but improperly, because it contains no argillaceous earth.

§ 129.

TERRA SILICEA (filiceous earth), united to *argillaceous* and a small quantity of *calcareous* earth.

CRONSTEDT Min. § 63. Petrofilex. Chert.

§ 130.

TERRA SILICEA (filiceous earth) united to *argillaceous* earth and a little *magnesia*.

CRONSTEDT Min. §66. *Feldspathum*. Feld spat.

§ 131.

TERRA SILICEA (filiceous earth), united to *magnesia*, mild *calcareous* earth, *fluor spar* and also to the *calxes* of *copper* and *iron*. Chrysoprasius. I have not examined this, but insert it upon the experiments of Mr. Achard.

To determine accurately the species of earths is the most difficult part of mineralogy, for innumerable analyses yet remain to be made. But that which now seems intricate and obscure will become plain and easy when experiments have been sufficiently multiplied.

CLASS III.
INFLAMMABLES,
OR
BITUMINA.

§ 132.

To this head we refer all fossils containing phlogiston in such great abundance, that under proper management they are inflammable. The Genera are obviously very few, and accurately speaking there is only one Genus. But since phlogiston is so very subtle as not by itself to become the object of our senses, it will perhaps be adviseable to consider its more simple combinations as Genera: this has long been done so far as respects the metals, by universal consent.

SULPHUR.

§ 133.

This name may be given to any acid coagulated by phlogiston into a solid form. If all metals consist of certain radical acids saturated with phlogiston, as is highly probable, and with respect to arsenic is indubitably proved; then metals ought to find a place here. But until this theory be established by numerous experiments, we shall only rank under this head the compounds which have not a metallic nature.

§ 134.

PHLOGISTON saturated with *vitriolic acid*.

CRONSTEDT, Min. § 151. *Common Brimstone*. Sulphur.

§ 135.

PHLOGISTON saturated with *aerial acid*.

CRONSTEDT Min. § 154. A. *plumbago*. Black-lead.

The true composition of this has been detected by Mr. SCHEELE.

§ 136.

PHLOGISTON united to the *acid* of *vitriol* and of *molybdæna*; or what amounts to the same, sulphur joined to the acid of molybdæna.

CRONSTEDT Min. § 154. b. c. *Molybdæna*. Molybdæna.

The acid of molybdæna has never yet been obtained quite free from phlogiston (§ 32). If this acid be of a metallic origin, molybdæna is a mineralized metallic substance, and should be placed with the other minerals.

PETROLEUM.

§ 137.

Phlogiston occurs also in the fossil kingdom, combined in an oily form; but many suppose this derived from the vegetable kingdom.

§ 138.

PETROLEUM pure and selected.

CRONSTEDT Min. §§ 147–150. Naptha. Rock oil.

§ 139.

PETROLEUM joined to *argillaceous* earth.

CRONSTEDT Min. §§ 157–160. *Lithantrax*. Pit Coal.

§ 140.

PETROLEUM united to *acid of amber*.

CRONSTEDT Min. §§ 133–146. *Succinum*. Amber.

Many contend that amber has a vegetable origin; but as the point is not very well determined; and as it is found amongst fossils, I still retain it here.

§ 141.

AMBERGRISE, according to the assertion of Mr. AUBLETT, is nothing more than the juice of a tree inspissated by evaporation into a concrete form. This tree grows in Guyana, and is called Cuma, but has not been investigated by any botanist. Pieces of this tree are said to be carried down into the rivers by heavy rains, and the specimens examined by Mr. ROUELLE had the odour and principal qualities of amber^[50]. RUMPHIUS, long since, mentioned a tree called Nanarium, whose juice resembled amber^[51].

DIAMOND.

§ 142.

At first sight I may seem to have acted erroneously, by separating this from the other gems, and inserting it here; but after due consideration, I know not where to place it better. It has never yet been decomposed by the liquid analysis^[52]; and when exposed to the fire in an open vessel, it is wholly consumed, burning with a lambent flame. This deflagration, though slow, shews decidedly its affinity to the inflammables: besides, in the focus of a burning glass, it leaves traces of soot^[53]. When further experiments teach us better, I shall willingly correct my error.

CLASS IV. METALS.

§ 143.

I have before mentioned the great affinity betwixt metallic and inflammable substances (§ 133). Zinc and arsenic stand, as it were, upon the borders betwixt them; for these, in proper circumstances, burn with a very evident flame. All the metallic substances contain phlogiston, and when, to a certain degree, deprived of it, fall into a powder like an earth; but their attractions for phlogiston are different. Most of them, when melted in a common way, and exposed to the air, have an earthy crust formed upon the surface, which cannot again be reduced to metal without the addition of some inflammable matter. The *base* metals, eleven in number, have this property: but the *noble* metals, platina, gold and silver, are so firmly connected to the phlogiston, that they never calcine under fusion, however long continued; and after being changed into a calx in the liquid way, when melted in the fire, they re-assume their metallic form, without any other phlogiston than what is contained in the matter of heat.

Quicksilver holds a kind of middle place; for, like the base metals, it may be calcined, though not readily; and like the noble ones, it may be reduced by heat alone.

I have placed each division of the metals in the order of their specific gravities.

Those metals, which are found in a perfect metallic state, are called *native*; those united to acids, or to sulphur, are said to be *mineralized*; and those which are only deprived of their phlogiston, *calcliform*^[54].

TABLE OF METALS.

METALS.		Specific Gravity.	Melting Heat^[55].	Saturating Phlogifton.	Attraction to faturating Phlogifton.
Gold		19,640	1301	394	1 or 2
Platina		21,000		756	1 or 2
Silver		10,552	1000	100	3
Quickfilver		14,110	−39 or −634	74	4
Lead		11,352	595	43	10
Copper		8,876	1450	312	8
Iron		7,800	1601	342	11
Tin		7,264	415	114	9
Bifmuth		9,670	494	57	7
Nickel	common	7,000	1301		
	pure	9,000	1601	156	11
Arfenic		8,308		109	5
Cobalt	common		1450		
	pure	7,700	1601		
Zinc		6,862	699	182	11
Antimony		6,860	809	120	6
Manganefe		6,850	very great	227	11

AURUM,
OR
GOLD.

§ 144.

The specific gravity of this metal, when pure, is 19,640. Aqua regia dissolves it; but except the dephlogisticated muriatic acid, and in certain circumstances the nitrous, no simple acid acts upon it, unless it has been previously calcined^[56]. The quantity of phlogiston necessarily taken away in the solution of 100 parts of gold, I estimate at about 394; whilst the same quantity of silver, loses by solution in the nitrous acid, 100^[57]. Gold retains the phlogiston necessary to its metallic form, more obstinately than any other metal, except, perhaps, platina. It melts and calcines in the focus of a burning glass at 1301 degrees of heat.

§ 145.

AURUM *nativum* (gold native) united to *silver*.

Native.

I do not know that gold has ever yet been found perfectly pure.

§ 146.

AURUM *nativum* (gold native) united to *copper*.

Native.

§ 147.

AURUM *nativum* (gold native) united to *filver* and *copper*.

Native.

§ 148.

AURUM *nativum* (gold native) united to *filver*, *copper*, and *iron*.

Native.

§ 149.

AURUM (gold), mineralized by *fulphur*, by means of iron.

Pyritical.

CRONSTEDT Min; § 166. a. *Pyrites aureus*.

But some doubt may be made about the mineralization of gold^[58].

§ 150.

AURUM (gold) mineralized by *fulphur*, together with *filver*, *lead*, and *iron*.

Minera aurifera Nagyayenlis.

I have not yet fully examined this^[59].

PLATINUM,
OR
PLATINA.

§ 151.

Its specific gravity is 18,000^[60], when very pure. It dissolves in aqua regia, and the loss of phlogiston during the solution, according to the experiments hitherto made may be expressed by 756. Besides the muriatic acid, which when dephlogisticated dissolves every metal, no acid acts upon platina without it has undergone a previous calcination. It seems to retain its phlogiston more obstinately than any other metal. To melt it requires a heat greater than that at which iron melts.

§ 152.

PLATINA *native* united to *iron*. Native.

CRONSTEDT Min. § 179.

I believe it has never been found quite free from iron, but this can be separated by art^[61].

ARGENTUM,
OR
SILVER.

§ 153.

Its specific gravity is 10,552. The nitrous acid readily dissolves it, the vitriolic must be boiling hot; the muriatic attracts its calx very strongly, but cannot remove its phlogiston and therefore cannot dissolve it in its metallic state. The quantity of this phlogiston which causes the difference betwixt its metallic and its calciform state I before expressed as 100 in 100 parts of silver. But the force with which it retains this portion of its phlogiston is less than that of gold; that is, it occupies the third place in a series of all the metals. It melts at 1000 degrees of heat.

§ 154.

ARGENTUM *nativum* (silver native) united to *gold*. Native.

§ 155.

ARGENTUM *nativum* (silver native), united to *copper*. Native.

§ 156.

ARGENTUM *nativum* (silver native), united both to *gold* and *copper*. Native.

§ 157.

ARGENTUM *nativum* (silver native), united to *iron*. Native.

§ 158.

ARGENTUM *nativum* (silver native), united to *arsenic*. Native.

The arsenic hardly exceeds $\frac{6}{100}$.

§ 159.

ARGENTUM *nativum* (silver native), united to *antimony*. Native.

When melted, it smokes but has no smell of arsenic.

§ 160.

ARGENTUM *nativum* (silver native), united to *arsenic* and *iron*. Native.

The three metallic ingredients are nearly in equal proportions.

All the species hitherto mentioned have metallic properties and appearances. The contaminating matters are sometimes extremely small, but not to be neglected when they exceed $\frac{1}{300}$ part of the mass.

§ 161.

ARGENTUM (silver) mineralized by the *vitriolic* and *muriatic* acids. Hornlike.

CRONSTEDT Min. §177. *Minera argenti cornea*. Horn-silver.

Mr. Woulfe^[62], detected the presence of the vitriolic acid. The silver seldom exceeds $\frac{70}{100}$. I know not whether it is ever altogether free from vitriolic acid.

§ 162.

ARGENTUM (filver), mineralized by the *vitriolic* and *muriatic* acids, and *fulphur*.

I doubt whether this be a distinct species, since the fulphur and the salts scarcely admit of any other than a mechanical union.

§ 163.

ARGENTUM (filver), mineralized by fulphur. Glassy.

CRONSTEDT Min. § 169. *Minera argenti vitrea*.

It sometimes contains $\frac{73}{100}$ of filver, or more.

§ 164.

ARGENTUM (filver), mineralized by *fulphur* and *iron*. Marcasitical.

CRONSTEDT Min. § 176, 10. *Pyrites argenteus*.

§ 165.

ARGENTUM (filver), mineralized by *fulphur* and *lead*. Potters.

CRONSTEDT Min. § 176, 8. *Galena*.

The filver is only a few half ounces in a hundred weight.

§ 166.

ARGENTUM (filver), mineralized by *fulphur* and *arsenic*. Red.

CRONSTEDT Min. § 170. *Minera argenti rubra*.

It contains about $\frac{70}{100}$ of filver. Iron is frequently present, as in most other species but not always.

§ 167.

ARGENTUM (silver), mineralized by *fulphur, arsenic, and iron*.
Glittering.

CRONSTEDT Min. § 172.

I have examined some specimens from Saxony which sometimes contain no silver. May we not therefore suppose that the silver is native and not mineralized?

§ 168.

ARGENTUM (silver), mineralized by *fulphur, arsenic, iron and cobalt*.
The silver is sometimes more than $\frac{50}{100}$.

§ 169.

ARGENTUM (silver), mineralized by *fulphur, arsenic, copper and iron*.
White ore.

CRONSTEDT Min. § 171. *Minera argenti alba*.

The proportion of silver varies much, sometimes it is $\frac{10}{100}$ or more.

§ 170.

ARGENTUM (silver), mineralized by *fulphur, arsenic, copper, iron, and antimony*. Grey ore.

CRONSTEDT Min. § 173. 6. *Minera argenti grisea*. In the province of Dal[\[63\]](#).

It contains $\frac{24}{100}$ of copper, feldom $\frac{5}{100}$ filver.

§ 171.

ARGENTUM (filver), mineralized by *fulphur*, *arfenic*, *antimony* and *iron*. Plumofe.

CRONSTEDT Min. §173. 5. *Federertz of the Germans*[\[64\]](#).

It feldom contains more than a few half ounces of filver in the hundred weight.

It is abfurd to found fpecies upon the differences of the matrix: thefe ought to be confidered elfewhere.

HYDRARGYRUM, OR QUICKSILVER.

§ 172.

Its specific gravity is 14,110. It has been erroneously ranked among the brittle metals, for at 654 degrees below 0 it freezes^[65], and then spreads under the hammer like lead. But as such an extreme degree of cold rarely happens unless artificially produced, we cease to wonder why it is always liquid or rather melted.

Nitrous acid dissolves it readily, vitriolic acid requires to be assisted by a boiling heat; muriatic acid does not act upon it all, unless previously deprived of as much phlogiston as in 100 parts may be called 74. The attractive power wherewith it retains this portion of phlogiston occupies the fourth place in the series; that is, it holds it less strongly than the noble but more strongly than the base metals.

§ 173.

HYDRARGYRUM *nativum* (quicksilver native). Native.

CRONSTEDT Min. § 217.

Whether it be entirely free from every metallic contamination I have not yet tried.

§ 174.

HYDRARGYRUM (quickfilver), united to *filver*. Amalgamated.

CRONSTEDT Min. § 217.

§ 175.

HYDRARGYRUM (quickfilver), mineralized by *muriatic* and *vitriolic* acids. Hornlike.

Mineralogy owes the discovery of this to Mr. WOULFE. Phil. Transf.

§ 176.

HYDRARGYRUM (quickfilver), mineralized by *fulphur*. Cinnabarine.

CRONSTEDT Min. § 218. *Cinnabaris*.

§ 177.

HYDRARGYRUM (quickfilver), mineralized by *fulphur* and *iron*. Martial.

I am doubtful whether this be a distinct species. The iron perhaps is only mechanically diffused.

§ 178.

HYDRARGYRUM (quickfilver), mineralized by *fulphur* and *copper*. Cuprous.

CRONSTEDT Min. § 219.

PLUMBUM,

OR

LEAD.

§ 179.

Its specific gravity is 11,352, greater than that of any other of the base metals. The nitrous acid perfectly dissolves it; the muriatic more difficultly; the vitriolic hardly at all, for the vitriol of lead being insoluble in water incrusts the metal, and prevents its solution. After calcination the weakest vegetable acids dissolve it, and acquire a sweet taste. The phlogiston necessary to be taken away in order that it may dissolve may be called 43, which is less than that of any other metal. Hence we understand why the calx of lead may be reduced with a very minute quantity of inflammable matter. With respect to the force wherewith it retains this phlogiston it occupies the tenth place. It melts at 595 degrees of heat.

§ 180.

PLUMBUM *nativum* (lead), though many mineralogists doubt whether it has ever yet been found. Native.

§ 181.

PLUMBUM (lead), mineralized by *vitriolic acid*. Vitriol of.

Originating from the decomposition of Galena. It is rarely met with. It was first observed by Mr. MONNET. It does not effervesce with acids. It may be reduced by the blowpipe upon charcoal.

§ 181*.

PLUMBUM (lead), mineralized by *vitriolic acid* and *iron*.

Existing in immense quantity in the island of Anglesea. It does not reduce with the blowpipe upon charcoal, but melts to a black glass^[66]. W.

§ 182.

PLUMBUM (lead), mineralized by the *acid of phosphorus*. Phosphorated.

This was discovered by Mr. GAHN. It does not effervesce with acids. It melts upon charcoal with the blowpipe, but is not perfectly reduced.

§ 183.

PLUMBUM (lead), mineralized by the *aerial acid*. Aerated.

CRONSTEDT Min. § 185.

It effervesces with acids, and is readily reduced upon charcoal^[67].

§ 184.

PLUMBUM (lead), mineralized by *fulphur*. Sulphurated.

CRONSTEDT Min. § 187.

§ 185.

PLUMBUM (lead), mineralized by *fulphur* and *silver*. Galena.

CRONSTEDT Min. § 188.

§ 186.

PLUMBUM (lead), mineralized *fulphur*, with *silver* and *iron*.

CRONSTEDT Min. § 189.

§ 187.

PLUMBUM (lead), mineralized by *fulphur*, with *silver* and *antimony*.
Radiated.

CRONSTEDT Min. § 190.

CUPRUM,
OR
COPPER.

§ 188.

Its specific gravity is 8,876. Nitrous acid dissolve it readily, muriatic acid flowly, and the vitriolic requires intense boiling. The phlogiston, separated in the solution of 100 parts, may be expressed by 312. The weakest vegetable acids act upon it, especially after calcination, and so do alkalies, the volatile alkaly especially. With respect to the power with which it retains the phlogiston, copper holds the eighth place. It melts with 1450 degrees of heat.

§ 189.

CUPRUM nativum (copper native). Native.

CRONSTEDT Min. § 193.

It's rarely found without some alloy of gold, silver or iron; but I have not yet fully examined it.

§ 190.

CUPRUM calciforme (copper), simply deprived of its phlogiston. Calciform.

CRONSTEDT Min. § 195.

§ 191.

CUPRUM (copper), mineralized by *muriatic acid* and *argillaceous earth*. Micaceous.

Mr. WERNER, in his translation of Cronstedt's Mineralogy, part 1, page 217, has described it accurately, and kindly sent me a specimen of it, which I analysed^[68].

§ 192.

CUPRUM (copper), mineralized by the *aerial acid*. Aerated.

CRONSTEDT Min. §§ 194, 196. b. 3.

Mr. FONTANA first pointed out its true composition. It contains about $\frac{2}{3}$ of copper, $\frac{1}{3}$ or $\frac{1}{4}$ of aerial acid, and a little water^[69].

§ 193.

CUPRUM (copper), mineralized by *fulphur*. Vitreous.

CRONSTEDT, Min. § 197. *Minera cupri vitrea*; a common, but improper name.

It generally contains some alloy of iron.

§ 194.

CUPRUM (copper), mineralized by *fulphur*, and a small proportion of *iron*.

CRONSTEDT Min. § 198, b. *Minera cupri lazurea*.

By a *small proportion* of iron, I mean less than the weight of the copper; by a large proportion, more. This contains from 40 to 50 per cent. of copper.

§ 195.

CUPRUM (copper), mineralized by *fulphur*, and a large proportion of *iron*. Pyritical.

CRONSTEDT Min. § 198. *Pyrites Cupri*.

The quantity of copper varies greatly, but seldom exceeds $\frac{40}{100}$.

§ 196.

CUPRUM (copper), mineralized by *fulphur*, *iron* and *arsenic*. Grey.

CRONSTEDT Min. § 198. a. *Pyrites cupri griseus*.

This frequently contains an alloy of silver. The copper rarely exceeds $\frac{60}{100}$.

FERRUM,

OR

IRON.

§ 197.

Its specific gravity is 7,800. All the acids readily dissolve it; but the vitriolic must be diluted, otherwise it may be boiled almost to dryness, without effecting it. The phlogiston, dislodged from centenary of ductile iron, may, as experiments now stand, be called 342; and this is so feebly retained, that this metal, with a few others, holds the eleventh, or lowest place in the series.

It requires an intense degree of heat to fuse it, viz. 1601, if the usual comparison betwixt the mercurial thermometer, and the metallic one of MORTIMER, be true. Iron is red hot at 1050 degrees of heat.

§ 198.

FERRUM nativum (iron) *native*. Native.

It can hardly be doubted, but that the great mass of iron, brought by PALLAS, from Siberia, into Europe, is the product of nature. Its composition resembles that of forged iron; for 100 parts of it yield, by means of the muriatic acid, 49 cubic inches of inflammable air; and from many experiments upon ductile iron, that is found to yield from 48 to 51^[70].

§ 199.

FERRUM nativum (iron) *native*, united to arsenic. Arsenical.

CRONSTEDT Min. § 243. B. *Mispickel*.

§ 200.

FERRUM (iron), with the power of attracting other iron. Loadstone.

CRONSTEDT Min. § 211. b. *Magnes*.

The cause of this property is yet unknown.

§ 201.

FERRUM (iron), with phlogiston enough to render it magnetic. Magnetic.

CRONSTEDT Min. §§ 212, 213.

But the quantity of phlogiston is far short of that which is necessary to render it ductile, for a centenary hardly contains more than three cubic inches of inflammable air.

§ 202.

FERRUM *calciforme* (iron calciform), simply deprived of phlogiston. Ochrous.

CRONSTEDT Min. §§ 202–206. *Bloodstone*.

§ 203.

FERRUM (iron), mineralized by *aerial acid*, *calcareous earth*, and *manganese*. White.

CRONSTEDT Min. § 20. *Minera ferri alba*.

§ 204.

FERRUM (iron), mineralized by *fulphur*. Pyritical.

CRONSTEDT Min. § 152. *Pyrites*.

§ 205.

FERRUM (iron) intimately united to a new brittle metal^[71], or to a peculiar modification of iron, rendering it brittle when cold. Cold-short.

In cold-short iron, a brittle metal exists, readily uniting to ductile iron, by the assistance of heat, but rendering it brittle when cold. This substance, dissolved in acids, forms Prussian blue with phlogisticated alkaly, but it is not magnetic: it affords a white calx, richer in phlogiston than the yellow calx of good iron.

I hope, by more experiments, soon to become better acquainted with it.

§ 206.

FERRUM *calciforme* (iron calciform), *phlogisticated* in a peculiar manner. Blue.

CRONSTEDT Min. § 208. *Cæruleum Berolinense nativum*.

Clay and mould are sometimes coloured superficially by a dilute blue, and sometimes the former, when newly dug up, is found to acquire this colour upon exposure to the air. It is evident that the basis of this colour is an irony matter, full of phlogiston; for, by ignition upon a charcoal fire, it flames, turns red, and becomes magnetic. With a gentle heat it becomes green, but when melted gives black scoriæ.

Alkalies, as well as acids, dissolve it, and the colour vanishes, but appears again, if precipitated from the former by acids, and from the latter by alkalies; but it has then a greenish cast, and soon becomes white. This white sediment, immersed in an infusion of galls, or of tea, recovers its former colour.

From what has been said, it appears that this colour, although analogous to the artificial Prussian blue, differs from it in its intensity, in the mode of its production, and in various properties. It keeps its colour in water, but turns black with oil.

STANNUM,

OR

TIN.

§ 207.

Its specific gravity is 7,264. Vitriolic, muriatic, acetous acids, and aqua regia, dissolve it, but the nitrous, especially when strong, attacks it so violently, that it soon reduces it to the state of an insoluble calx.

The quantity of phlogiston it loses by solution, may be called 114; and this it retains with a force that gives it the ninth place in the series. It melts easier than any metal, except quicksilver, viz. at 415 degrees.

§ 208.

STANNUM *nativum* (tin). Native.

This I have not seen. Some doubts are entertained of its true nature, and, perhaps, not without reason.

§ 208*.

STANNUM *fulphuratum* (tin), mineralized by *fulphur*. Sulphurated.

[See the [Preface](#).]

§ 209.

STANNUM *calciforme* (tin) calciform, contaminated by *iron*. Calciform.

VISMUTUM,
OR
BISMUTH.

§ 210.

The heaviest of all the brittle metals that follow it, its specific gravity being 9,670. Nitrous acid, and aqua regia dissolve it perfectly. The vitriolic acid must be boiled nearly to dryness before it acts upon it, and the muriatic acid only attacks its calx. The quantity of phlogiston which resists the action of menstrua, is expressed by 57; and its power of retaining it ranks it in the seventh place. It melts at the heat of 494 degrees.

§ 211.

VISMUTUM *nativum* (bismuth). Native.

CRONSTEDT Min. § 222.

§ 212.

VISMUTUM *calciforme* (bismuth). Calciform.

CRONSTEDT Min. § 223.

I am not able to say whether this is merely deprived of its phlogiston, or whether it is not also mineralized by aerial acid.

§ 213.

VISMUTUM (bismuth) mineralized by *fulphur*. Sulphurated.

CRONSTEDT Min. § 224.

§ 214.

VISMUTUM (bismuth) mineralized by *fulphur* and *iron*. Pyritical.

CRONSTEDT Min. § 225.

NICCOLUM,

OR

NICKEL.

§ 215.

The regulus, when depurated, has a specific gravity of 9,000, or more; but the common regulus, obtained by the first reduction, little exceeds 7,000. Aqua regia, and nitrous acid, dissolve it perfectly; muriatic acid, slowly; vitriolic acid, not without boiling almost to dryness and the acetic acid does not act upon it, unless in a calciform state. The quantity of phlogiston separated by solution, may be called 156; and this it retains with a force about equal to that with which iron retains its phlogiston (§ 197).

The heat necessary to melt it, is about equal to that which gold requires; but when depurated, it is almost as difficult to melt as iron.

The properties of it are more fully examined elsewhere^[72].

§ 216.

NICCOLUM *nativum* (nickel) native, united to *iron* and *arsenic*. Native.

It sometimes, perhaps, contains cobalt. As it contains neither sulphur nor mineralizing acid, and is perfectly in its metallic form, it must be called *native*, although joined to other metals.

§ 217.

NICCOLUM *aeratum* (nickel) mineralized by *aerial acid*. Aerated.

CRONSTEDT Min. § 255.

§ 218.

NICCOLUM (nickel) mineralized by *fulphur, arfenic, cobalt, and iron*.
Mineralized.

CRONSTEDT Min. § 256. Cuprum Nicolai. Kupfer nickel.

ARSENICUM,

OR

ARSENIC.

§ 219.

The specific gravity of the radical acid, is 3,391; of white arsenic, 3,706; of its glassy state, 5,000; and its regulus, 8,308. Aqua regia, and muriatic acid, dissolve it perfectly; the vitriolic acid requires boiling; the acetous acts only upon its calx: the nitrous acid not only takes away as much phlogiston as may be expressed by 109, deprived of which the regulus is reduced to the state of a calx, but in a large quantity, assisted by a proper degree of heat, it at length so far dephlogisticates this calx, as to leave the acid of arsenic alone. These phænomena are well worthy of observation, as they seem to lay open the nature of metals in general. From analogy, it is probable that every metal contains a radical acid of a peculiar nature, which, with a certain quantity of phlogiston, is coagulated into a metallic calx; but with a larger quantity, sufficient to saturate it, forms a complete metal. The radical acid retains the coagulating phlogiston much more strongly than that which is necessary to the saturation. But different metallic acids retain both with different degrees of attraction. Hence the noble metals cannot be calcined in the dry way; it is only by acid menstrua that they can be brought into that form; but all the others lose their saturating phlogiston in the fire, though with more or less difficulty. I have distinctly observed eleven different degrees of resistance: thus, gold may be precipitated by all the other metals, except perhaps platina, which I think may thus be explained. The calx of gold having the greatest attraction for phlogiston, takes it from all other metals, and thus losing its solubility falls down in a metallic state. Therefore gold in the series of metals, occupies at least the second place. Platina is

precipitated by all, but less evidently than gold. To this therefore, I think we must give the first place, and so on of the others as I have remarked in the character of each metal. As nickel, cobalt, iron, manganese and zinc, do not precipitate one another, they are put together in the last and eleventh place^[73].

In order to obtain the radical acids we must separate them from the coagulating phlogiston. If the industry of chemists ever effects this, I am confident that metallurgy will be wonderfully elucidated. This therefore is a task to which our labours must be directed. I know that analogy must be cautiously trusted, but it at least leads us to new experiments. Hitherto this operation has only succeeded with arsenic; and it is worth notice, that this metal which holds the fifth place with respect to its quantity of phlogiston, should be inferior to all others with regard to the attraction by which the coagulating quantity is retained.

Arsenic melts, but the moment it suffers heat enough to melt it, it volatilizes, unless it be first calcined. The regulus thrown upon a plate of iron properly heated, presently takes fire and calcines, diffusing a smell like garlic^[74].

§ 220.

ARSENICUM *nativum* (arsenic), native, united to iron. Native.

CRONSTEDT Min. § 239.

I have never found it free from martial impregnation.

§ 221.

ARSENICUM *nativum* (arsenic), native, united to silver.

§ 222.

ARSENICUM *calciforme* (arsenic), deprived of phlogiston. Calciform.

CRONSTEDT Min. § 240.

§ 223.

ARSENICUM (arfenic), mineralized by *fulphur*. Yellow.

CRONSTEDT Min. § 241. *Auripigmentum. Rifiigallum.*

§ 224.

ARSENICUM (arfenic), mineralized by *fulphur* and *iron*. Pyritical.

CRONSTEDT Min. § 243. A. *Pyrites arfenicalis.*

COBALTUM,

OR

COBALT.

§ 225.

Its specific gravity is 7,700. Nitrous acid and aqua regia readily dissolve it. The vitriolic acid requires to be boiled nearly to dryness. The muriatic and acetous acids do not act upon it unless previously calcined. 270 expresses the quantity of saturating phlogiston, which it retains with the same force that iron does. Common regulus melts in the same heat that copper does, but when well purified it is hardly easier to melt than iron.

§ 226.

COBALTUM *nativum* (cobalt), native and united to *arsenic*. Native.

CRONSTEDT Min. § 249.

§ 227.

COBALTUM *calciforme* (cobalt). Calciform.

CRONSTEDT Min. § 247.

It is found variously mixed, principally with arsenic, iron and copper, but whether mechanically or by a more intimate union I know not.

§ 228.

COBALTUM (cobalt), mineralized by acid of *arsenic*. Red.

CRONSTEDT Min. § 248.

The small specimens that I have been able to examine point out such a composition^[75].

§ 229.

COBALTUM (cobalt), contaminated by *iron* and *vitriolic acid*. Vitriolic.

CRONSTEDT Min. § 250.

§ 230.

COBALTUM (cobalt), mineralized by *fulphur*, *arsenic* and *iron*. Glanz-cobalt.

CRONSTEDT Min. § 251.

§ 231.

COBALTUM (cobalt), mineralized by *fulphur*, *arsenic*, *iron* and *nickel*. Kupfernickel.

CRONSTEDT Min. § 252.

ZINCUM

OR

ZINC.

§ 232.

Its specific gravity is 6,862. All the acids dissolve it readily and with effervescence, which denotes its very lax union with the inflammable principle, as was remarked before (§ 219). 182 expresses the quantity of phlogiston it loses in solution. It melts in a heat of 699 degrees; and if the heat be a little increased it takes fire; and dissipates in white flowers^[76].

§ 233.

ZINCUM *calciforme* (zinc), calciform simply deprived of its phlogiston. Calciform.

CRONSTEDT Min. § 228. A. *Lapis calaminaris*.

It is almost always mixed with clay or calciform iron.

§ 234.

ZINCUM (zinc), mineralized by *aerial acid*. Aerated.

CRONSTEDT Min. § 228. A. 1.

§ 235.

ZINCUM (zinc) with *aerial acid* and mixed with *filiceous* matter. Siliceous.

D. A. BORN sent me crystals of this species, which exposed to the fire gave out aerial acid, but they were not wholly soluble in acids.

§ 236.

ZINCUM (zinc), mineralized by *fulphur* and *iron*. Black jack.

CRONSTEDT Min. §§ 229. 230. *Pseudogalena*.

ANTIMONIUM

OR

ANTIMONY.

§ 237.

Its specific gravity is 6,860. Aqua regia dissolves it well; vitriolic acid requires boiling; muriatic and acetic acids act hardly at all upon it, unless previously calcined. The nitrous acid corrodes it so as to prevent the solution. The phlogiston it loses in solution is expressed by 120, and with respect to the force wherewith it retains this, it stands in the sixth place. It melts at a heat of 809 degrees.

§ 238.

ANTIMONIUM nativum (antimony). Native.

CRONSTEDT Min. § 238.

§ 239.

ANTIMONIUM (antimony), mineralized by sulphur. Sulphurated.

CRONSTEDT Min. § 234.

§ 240.

ANTIMONIUM (antimony) mineralized by sulphur and arsenic. Red.

CRONSTEDT Min. § 235.

MANGANESIUM,
OR
MANGANESE.

§ 241.

Its specific gravity is 6,850. This new metal is soluble in all the acids, and is so readily deprived of its saturating phlogiston that with iron and some others it stands the lowest in the series. 227 expresses the quantity of phlogiston it loses in solution. It is very difficult to melt, more so than iron.

§ 242.

MANGANESIUM *calciforme* (manganese) simply deprived of phlogiston. Calciform.

CRONSTEDT Min. § 114.

§ 243.

MANGANESIUM (manganese) mineralized by *aerial acid*.
Aerated.

CRONSTEDT Min. § 115. 1. a.

APPENDIX

THE FIRST.

§ 244.

In the preceding pages only the more simple combinations occur, whose principles are either chemically united or at least so subtly interwoven that the texture appears perfectly homogeneous. But if two or more of these species, forming little distinct masses are cemented together, these mechanical mixtures, discernible by the eye ought to constitute a new series, to be distinguished by their component parts as the others were by their first principles or chemical elements. Such compositions may well be excluded from the present work, but upon account of their extensive physical, æconomical and metallurgical uses, I propose to give a flight sketch of them here, enumerating the more remarkable Genera.

§ 245.

In a general view it appears that not only several species cemented together may be referred to this place, but likewise those which are mechanically diffused in a powdery or an earthy form.

§246.

From the laws of combination it is evident, that according to the arrangement of fossils into four classes, there can be only TEN Genera composed of two, FOUR of three, and ONE of four constituent parts. And although so many have not yet been detected, yet it is better to mention them here as the industry of a future age will probably discover more. The

species are formed from the differences of the more simple species and their component parts.

Salts *with* Salts.

§ 247.

This composition can hardly ever constitute a genus, if it must be made in a dry and concrete form; for excepting gypsum, the other native salts readily dissolve in water, and by evaporation are so mixed together as not readily to be discerned by the eye. Yet the fossil alkali mixed with common salt will perhaps find a place here. The contents of mineral waters may likewise be referred here, since every material difference in them depends upon the particles dissolved.

Salts *with* Earths.

§ 248.

This mixture is hardly to be found but where bits of gypsum are concreted to matters of an earthy nature.

Salts *with* Inflammables.

§ 249.

May perhaps be found in volcanoes.

Salts *with* Metals.

§ 250.

If gypsum forms the matrix of any metal, it must be placed here.

Earths *with* Earths.

§ 251.

To this head belong most of the *faxa* (stones), enumerated by Mr. Cronstedt, which form the immense bulk of mountains, and deserve our particular attention, in order that, being better acquainted with the nature and structure of the shell of the earth, we may be able to point out the coverings of minerals, and convert them all to our use.

Earths *with* Inflammables.

§ 252.

Lumps of mountain pitch are frequently connected with stones, and sulphureous matters are found diffused through earthy materials.

Earths *with* Metals.

§ 253.

This genus contains the peculiar *matrices* of *metals*, a judicious confideration of which would be particularly ufeul to miners.

Inflammables *with* Inflammables.

§ 254.

Perhaps, in some places, sulphureous matters are found mixed with mountain pitch.

Inflammables *with* Metals.

§ 255.

If plumbago (black-lead) or common sulphur, shall ever be found mixed with metallic substances, such species must stand under this genus.

Metals *with* Metals.

§ 256.

We know that some metals, in the bosom of the earth, are almost always mixed, whilst others are rarely, or never, found together. A more accurate knowledge of these things, would illustrate physical geography, as well as metallurgy.

We now proceed to the more compound genera.

Salts *with* Earths *and* Inflammables.

§ 257.

This genus can hardly ever occur but in countries formerly exposed to subterranean fires. [\[77\]](#)

Salts *with* Earths *and* Metals.

§ 258.

To be expected amongst volcanic productions.

Salts *with* Inflammables *and* Metals.

§259.

To be fought for in the productions of volcanoes.

Earths *with* Inflammables *and* Metals.

§ 260.

Obvious amongst the productions of volcanoes, otherwise extremely rare.

Salts *with* Earths, Inflammables, *and* Metals.

§ 261.

Hardly to be expected but in volcanic mountains.

APPENDIX THE SECOND.

§ 262.

Fossils externally resembling animals or vegetables, originate from foreign matters, which by some peculiar process are changed in the bosom of the earth, or are so impregnated by mineral particles gradually occupying the place of those which have putrified, that they no longer resemble organic substances, except in figure.—These are commonly called PETREFACTIONS.

§ 263.

The harder shells of animals exposed to the weather, are not always exempt from destruction; for their gelatinous matter being gradually destroyed by putrefaction, they become brittle, and in a manner calcined. In less exposed situations, some of them preserve the nature of their materials, but acquire a spar-like texture.

§ 264.

We must carefully distinguish betwixt the foreign bodies themselves, changed or petrified, and their impressions upon the surrounding matrices. Sometimes the body is entirely destroyed, forming a cavity in the surrounding matter, and this cavity afterwards is filled with other materials. Nuclei, or kernels, are likewise found, formed within the cavities of the harder shells, and bearing the form of their internal surface.

§ 265.

I am far from thinking the knowledge of petrefactions is barren and useless. We may, and ought, to consider them as medals deposited by the hand of nature, in memory of the more remarkable changes on the surface of the earth, and from which the time and order of the work may, in some measure, be judged of, whilst other monuments are silent. These, being properly interpreted, shew us their native situations in the former state of the surface of the earth, and teach us the unbounded empire of the sea, and the consequent changes. By them we learn to distinguish the ancient and modern foundations of the mineral kingdom; for those which are not formed of petrefactions, and never contain them, are doubtless of greater antiquity than animals or vegetables; and, lastly, by their figure they shew us the inhabitants of our globe, especially those of the greatest depths of the ocean.

§ 266.

Mr. CRONSTEDT has admirably arranged the petrefactions; we think it right, therefore, to retain his method. The Genera are built upon the Genera of fossils, and arranged like the four classes thereof; the species upon their species, and the varieties upon the organic substances that have been changed. The following are the Genera hitherto discovered.

Saline Calcareous Earth with an organic Form.

§ 267.

Gypseous petrefactions are very rare.

Saline Iron with an organic Form.

§ 268.

Human bodies have sometimes been found indurated and penetrated by vitriol of iron; so likewise have plants, their roots especially. In the open air they moulder away.

Mild Calcareous Earth with an organic Form.

§ 269.

This constitutes the substance of most petrefactions.

Clay with an organic Form.

§ 270.

It is remarkable, that petrefactions found in clay are compressed, although, in subjacent calcareous strata, they preserve their natural figure. Similar compressed petrefactions are also found in the marly schistus.

Siliceous Earth with an organic Form.

§ 271.

Siliceous petrefactions are sometimes met with, but, in general, this material forms only nuclei (§ 264). Trunks of trees are sometimes found changed into agate. The celebrated FERBER has seen petrefactions in chert and jasper, and the illustrious BORN mentions corallines (porpitæ) in finople or martial jasper.

Earth organic.

§ 272.

Animals and vegetables are resolved by putrefaction into an earth, which may be regarded as forming a peculiar genus, until every appearance of organization being obliterated, at length it comes to be considered as common earth.

Petroleum impregnating organic Bodies.

§ 273.

Wood, penetrated by indurated petroleum, forms a remarkable variety of coal.

Silver with an organic Form.

§ 274.

Native silver is sometimes inherent in petrefactions, but never, to my knowledge, constitutes the substance of them, unless mineralized with copper and sulphur.

Quicksilver in an organic Form.

§ 275.

When mineralized by sulphur, it sometimes, though very rarely, constitutes petrefactions.

Copper with an organic Form.

§ 276.

Bones and teeth are sometimes found replete with the blue calx of copper. Bits of copper pyrites often stick in petrefactions, but seldom constitute their whole substance. I have some such from Norway, in a matrix of magnetical iron ore.

Iron with an organic Form.

§ 277.

Californian iron sometimes is found in the shape of roots and branches of trees. When mineralized by sulphur, it frequently exists in petrefactions, but seldom constitutes the whole mass.

Zinc with an organic Form.

§ 278.

I have seen pseudo-galena (black jack), in the form of coral.

§ 279.

Some modern writers, as well as Mr. CRONSTEDT, place the productions of *volcanoes* in an appendix by themselves; but, I think, to no good purpose. Things formed by the hand of nature, whether by a liquid or a dry process, must not be disjoined; for she frequently avails herself of both methods in one and the same instance. And, indeed, the origin of many things is so very doubtful, every vestige thereof being obliterated, that even an Œdipus could not with certainty determine how they were produced. And, on the other hand, many assert, that almost the whole of the mineral kingdom is the product of fire. To avoid error, therefore, it is better to class fossil substances according to their constituent parts, which proper experiments will lay open to us; for we can seldom know their origin or formation.

Homogeneous substances joined together, but not primitive, will find a place among the stones, or elsewhere, in the first appendix.

FINIS.

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ERRATA.

Page [9](#), n. for Dr. WERNER, read Profeffor WERNER.

[15](#), line 23, for VANDELL, read VANDELLI.

[21](#), line 10, for red, read green.

[41](#), line 9, for acidor, read acid or.

[71](#), line 5, for 27,500, read 21,000.

[75](#), note — 27,500, read 21,000.

[95](#), line 1, for 100 parts, read centenary.

[97](#), line 7, for berolineufe read, Berolinenfe.

[1](#). In this tranflation they are introduced in their proper places. W.

[2](#). There is no difficulty in doing this: either the foſſil, or the vegetable fixed alkaly phlogifticated, precipitate the terra ponderoſa, inſtantly and entirely, out of the nitrous, muriatic, or vegetable acids. W.

[3](#). Opufc. chemica, vol. II. page 2–10.

[4](#). Conſult particularly Profeffor WERNER’S treatiſe on the *external characters of foſſils* printed in German in the year 1774.

[5](#). The latter part of this definition does not apply perfectly well to ſome of the ſimple ſalts. I ſhall therefore offer another, given by Dr. Cullen, viz. “*Saline bodies* are ſapid, miſcible with water, and not inflammable.” I am ſenſible too that this definition is not perfectly unexceptionable, ſince it has been found that vol. alkaly in an aerial ſtate is in a certain degree inflammable. W.

[6](#). As the tincture of heliotropium is the nicelt known teſt of the preſence of an acid, it may not be amiſs to mention that it may be had from dyers under the name of litmus. It is very cheap, and generally requires to be greatly diluted with diſtilled water before it can be uſed. W.

[7](#). De thermis pativinis.

[8.](#) The most highly coloured and fuming nitrous acid may readily be rendered colourless by boiling it hastily in an open vessel. Part of the acid flies off, carrying the superabundant phlogiston along with it, in the form of nitrous air. W.

[9.](#) N. Acta Upf. vol. II. p. 202.

[10.](#) M. Margraaf.

[11.](#) I have some reason to believe that the Nevil Holt water does contain some of this acid in an uncombined state. W.

[12.](#) Opuscul: vol. II. p. 40.

[13.](#) Called Derbyshire fluor; Cornish fluor, blue John. W.

[14.](#) D. SCHEELE Act. Stockh. 1778.

[15.](#) It has been lately obtained in great abundance from bones. W.

[16.](#) Opusc. chem. vol. II. p. 424.

[17.](#) De Sale sedativo naturali, 1778.

[18.](#) It is found in a separate state in large quantities in some of our mines and wells, and is called the *choak damp*. In the famous Grotto del Cano too it exists tolerably pure. W.

[19.](#) D. D. MARGRAAF, WEIGLEB.

[20.](#) Opusc. chem. vol. II. p. 368.

[21.](#) MARGRAAF Opusc.

[22.](#) CAVENDISH Phil. Trans. 1767.

[23.](#) Dr. Home, in his essay on bleaching, says it is found in coal mines in this island, and a friend assures me that he has obtained it from the water issuing out of coal pits. W.

[24.](#) As volatile alkali may be obtained in large quantities from pit coal, and produced by processes not dependant upon putrefaction, there is reason to believe that the vitriolic ammoniac may be formed in several ways not noticed by the author. W.

[25.](#) Acta Stockh. 1772.

[26.](#) From some experiments lately made I found that both tinkal and purified borax, required twice their weight of sedative salt, to neutralize them perfectly so that they would no longer change vegetable blues to a green. W.

[27.](#) Baumé mem. des sc. etr. tom. iv.

[28.](#) Phil. Trans. 1767.

[29.](#) HENCHEL Bethesda port.

[30.](#) BOMARE Dictionaire.

[31.](#) I have lately discovered a specimen of *TERRA PONDEROSA aerata* got out of a mine in this kingdom. It is very pure, and in a large mass. As this substance is a new acquisition to mineralogy, and may be turned to useful purposes in Chemistry, I intend shortly to present a more particular account of it to the Royal Society. W.

[32.](#) Conf. Præl. SCHEFFERI, § 188, not. 2.

[33.](#) Margraaf Kl. Schrift. tom. II. p. 191.

[34.](#) I found it in considerable quantity in the Nevil Holt water, when I analyzed it six years ago; and it is probable that the Ballycastle water in Ireland, likewise contains it. W.

[35.](#) In the original the word is MAGNESIUM, but it is here changed, by the advice of Dr. Swedjar and the concurrence of professor Bergman to MANGANESIUM, in order to prevent confusion from its similarity to Magnesia. W.

[36.](#) Mr. MONNET de aquis mineralibus.

[37.](#) Opusc. chem. vol. I. p. 394–399.

[38.](#) The author speaks here of such as he obtained by precipitation from acids, but the *native* *TERRA PONDEROSA aerata* (see note at page [28](#)) has a specific gravity of nearly 4, 338. W.

[39.](#) Opusc. vol. i. p. 21, 398.

[40.](#) N. Acta Upf. Vol. II. page 198.

[41.](#) Opusc. chem. vol. I, page 23.

[42.](#) Opusc. chem. vol. I, page 398.

[43.](#) Opusc. chem. vol. II. p. 29, 373.

[44.](#) Professor *Bergman* does not here seem to be sufficiently aware of the difference between our Devonshire pipe clay, and that which is used in the manufacture of porcelain. The former, in an open fire, burns to a blueish grey, or pigeon colour; the latter remains white. The former seems to be the same as the Cologne and Maestricht pipe clay, of Cronstedt, §78; the latter is a decayed Feldspath, and consequently, according to our author, (§ 130) contains magnesia. Our porcelain clay, likewise, has quartz, crystals, and mica mixed with it, parts of the granite which it originally composed. Before it is used the quartz is separated, but the mica remains. I am indebted to my friend Mr. Watt for these observations. W.

[45.](#) I have taken the liberty to add this species upon our author's own authority. See BERGMAN Diff. de Lithomarga, page 13.

[46.](#) N. Acta Upfal. vol. III, page 121.

[47.](#) Opusc. vol. I, page 291, 292.

[48](#). It is probable, that in another edition, the author may see reason to separate the *mica* from the *talc*; as some experiments I have made, though yet too imperfect for publication, seem to indicate the necessity of such a measure. W.

[49](#). Opusc. vol. ii. p. 49.

[50](#). Hift. des Plantes de la Gujane. 1774.

[51](#). Dr. SWEDIAR lately presented a paper to the Royal Society, from which it appears highly probable that Ambergrife is nothing but the indurated faeces of the Sperma Ceti whale, who feeds upon the cuttle fish. He has found the beaks of that fish intermixed with the ambergrife, in the form of black spots. W.

[52](#). Opusc. Vol. II, page 112.

[53](#). Lavoisier, Mem. de l'Acad. de Paris.

[54](#). Opusc. vol. II. page 275.

[55](#). The degrees of heat here expressed, are according to Fahrenheit's scale.

By *saturating phlogiston*, Professor Bergman means to express the proportionate quantities taken away from each metallic substance, when dissolved by means of acids, and of course reduced to a calciform state. The last column only expresses their attractions to this part of their phlogiston, not to that which still remains united to them in a calciform state. W.

[56](#). Opusc. Vol. II. page 374–376.

[57](#). Differtatio de quantitate Phlogisti in diversis metallis.

[58](#). Opusc. chem. vol. II, page 411.

[59](#). Opusc. chem. Vol. II, page 413.

[60](#). From some late experiments made upon platina by the Count de SIKENGEN, and published in German by professor Succow, it appears that the specific gravity of pure platina is 21,000. When perfectly pure and in its metallic state it was not calcined by deflagration with nitre, it did not admit of being hardened or softened by tempering, like steel or other metals; it was drawn into a wire $\frac{1}{1940}$ of a line in diameter; this wire admitted of being flattened, and had more strength than a wire of gold or silver of the same size. This platina is not fusible by the strongest fire, but melts in the focus of a burning glass; its colour white, shining like fine silver.

From considering the very interesting experiments of the Count de Sikengen, I apprehend the following method to obtain pure and malleable platina will be found a good one.

Dissolve the grains of native platina that are least magnetic, in aqua regia. Precipitate the iron by means of phlogisticated fixed alkali. Then precipitate whatever else will fall, by caustic vegetable alkali. Saturate the liquor with caustic fossil alkali, and set it by to crystallize. The yellow crystals thus obtained are to be hammered together at a welding heat, and the metallic parts will unite. W.

[61](#). Opusc. chem. vol. II, page 181.

[62](#). Phil. Transl.

[63.](#) This reference is not to be found in the English edition of Cronstedt. I imagine it should be § 174. 6. where it is called the Dal Falertz. W.

[64.](#) In this reference too I suspect a mistake. It ought I believe to be 173, 6. W.

[65.](#) Some late experiments made at Hudson's Bay seem to prove that Quicksilver congeals and becomes malleable at 39 degrees below 0. See Lond. Med. Journal, page 205, for the year 1783. W.

[66.](#) When I introduce a new *species* I repeat the preceding number, with the addition of an afterisk, rather than break in upon the order of the author's numbers. I intend shortly to publish an exact analysis of this substance. W.

[67.](#) Opusc. chem. vol. II, page 426.

[68.](#) Opusc. vol. II. page 431.

[69.](#) Opusc. chem. vol. II. p. 429.

[70.](#) Diff. de Analyfi. ferri.

[71.](#) Called Sideritis, from its resemblance to iron. W.

[72.](#) Opusc. chem. vol. II. p. 231.

[73.](#) Diff. de quantitate phlogisti in metallis.

[74.](#) Opusc. chem. vol. II, p. 272.

[75.](#) Opusc. chem. vol. II, p. 446.

[76.](#) Opusc. Vol. II, page 309.

[77.](#) Some of the sulphur and alum, sublimed by the subterranean fires near Bilston, contain filiceous earth. W.

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Page	Changed from	Changed to
50	of 2,155. 100 parts of contain about 25 of	of 2,155. 100 parts of it contain about 25 of

- Typos fixed; non-standard spelling and dialect retained.
- Corrected [Errata](#).
- Used numbers for footnotes, placing them all at the end of the last chapter.

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